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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Time: 03 Hours

Code: 100204 ***Subject: Elements of Mechanical Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) Which of the following is a renewable source of energy?
- (i) Coal
- (ii) Petroleum
- (iii) Solar energy
- (iv) Natural gas

(b) Which gas is primarily produced in a biogas plant?
- (i) Carbon dioxide
- (ii) Methane
- (iii) Hydrogen
- (iv) Nitrogen

(c) Which of the following is an extensive property of a thermodynamic system?
- (i) Pressure
- (ii) Temperature
- (iii) Volume
- (iv) Density

(d) Which of the following statements is TRUE about internal energy?
- (i) It depends only on pressure
- (ii) It depends only on temperature
- (iii) It depends on both temperature and molecular structure
- (iv) It is independent of state

(e) Which of the following represents a cyclic process?
- (i) The system returns to its initial state after the process
- (ii) The system undergoes irreversible changes
- (iii) Heat transfer is zero
- (iv) Work transfer is zero

(f) In a vapour compression refrigeration cycle, the refrigerant absorbs heat in which component?
- (i) Condenser
- (ii) Compressor
- (iii) Evaporator
- (iv) Expansion valve

(g) Which of the following is the main advantage of vapour absorption refrigeration over vapour compression refrigeration?
- (i) Lower initial cost
- (ii) No need for heat source
- (iii) Uses mechanical energy instead of thermal energy
- (iv) Can operate without electricity (uses thermal energy)

(h) In a simple steam turbine, the energy conversion primarily occurs from
- (i) Heat energy to mechanical energy
- (ii) Mechanical energy to heat energy
- (iii) Electrical energy to mechanical energy
- (iv) Chemical energy to electrical energy

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(i) In a thermal power plant, the working fluid used in the Rankine cycle is

(i) Air
(ii) Steam/Water
(iii) Water only
(iv) Gasoline

(j) Which of the following is a ferrous metal?

(i) Copper
(ii) Aluminum
(iii) Cast Iron
(iv) Brass

Q.2 Solve both questions : **[14]**

(a) Describe biogas and biomass energy. Mention their merits, demerits, and major applications.

(b) Explain the origin and forms of renewable energy sources such as solar and wind energy.

Q.3 Solve both questions : **[14]**

(a) What are non-renewable or conventional sources of energy? Explain fossil fuels with examples.

(b) Compare renewable and non-renewable energy sources in terms of availability, environmental impact, and sustainability.

Q.4 Solve both questions : **[14]**

(a) Define a thermodynamic system. Explain its types with suitable examples.

(b) Explain the concepts of state, process, and cycle in thermodynamics with diagrams.

Q.5 Solve both questions : **[14]**

(a) Explain the principle of operation of a vapour compression refrigeration system with the help of a neat sketch.

(b) Compare vapour absorption and vapour compression refrigeration systems in terms of energy source efficiency and applications.

Q.6 Solve both questions : **[14]**

(a) Explain the working principle of a simple steam turbine with a neat sketch.

(b) Compare SI (Spark Ignition) and CI (Compression Ignition) engines in terms of fuel used, ignition method, and applications.

Q.7 Solve both questions : **[14]**

(a) Describe the main components and working of hydel power plant.

(b) Discuss the principle of nuclear power generation. What are its advantages and disadvantages compared to thermal power plants?

Q.8 Solve both questions : **[14]**

(a) Define ferrous and non-ferrous metals with examples.

(b) Explain the mechanical properties of engineering materials such as strength, hardness, and resilience with suitable examples.

Q.9 Solve both questions : **[14]**

(a) What is quenching? Discuss the types of quenching media and their influence on steel hardness.

(b) Define case-hardening. Explain its methods and applications in engineering components.

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